

Dr.Rammanohar Lohia Avadh University , Ayodhya

Institute of Engineering and Technology

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING



Presented by:

Vaibhav Tiwari

Roll No. 22157

Suraj Verma

Roll No. 22151

Pradeep Maurya

Roll No. 22127

MULTILINGUAL LANGUAGE DETECTION AND TRANSLATION USING AI

Presented To:

Dr.Ashish Kumar Pandey

Assistant Professor

(CSE)

Multilingual Language Detection And Translator Using AI

PYTHON • ARTIFICIAL INTELLIGENCE • NLP

Breaking language barriers instantly using Natural Language Processing and Machine Translation, built entirely in Python

A Mini / Major Project Presentation



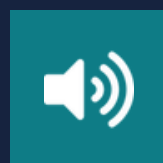
Voice Input



Detection



Translation



Speech Output

Introduction

Why do we need a real-time translator?

- The world is more connected than ever, but language remains one of the biggest barriers to communication, business and travel.
- Manual translation is slow, requires internet look-ups, and doesn't scale for live conversations, chats or customer support.
- This project builds a Python-based system that automatically detects the language of an input (text or voice) and instantly translates it, with optional speech output.

THE LANGUAGE BARRIER — IN NUMBERS

7,000+

Spoken languages exist worldwide

100+

Languages supported by modern translation engines

< 1 sec

Target detection + translation latency for a smooth chat-like experience

Project Objectives



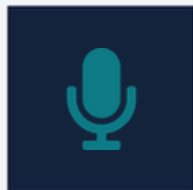
Automatic Language Detection

Identify the language of any typed or spoken input with high accuracy, without the user selecting it manually.



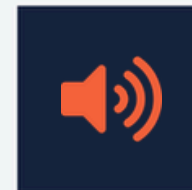
Instant Translation

Translate the detected text into a target language in real time using a machine translation engine.



Speech-to-Text Input

Allow users to simply speak into a microphone instead of typing, using speech recognition.



Text-to-Speech Output

Convert the translated text back into natural-sounding speech in the target language.

System Architecture

End-to-end pipeline from user input to translated output



User Input

Text or Voice



Preprocessing

Clean & tokenize text



Language Detection

langdetect / langid



Translation Engine

deep-translator API



Text-to-Speech

gTTS / pyttsx3



Output

Translated text + audio

The pipeline runs continuously: as soon as input is captured, it flows through detection and translation modules and returns an output within a second — enabling near real-time conversation.

Technology Stack

Built entirely using Python and its powerful open-source libraries



langdetect / langid

Detects the language of input text using statistical n-gram models



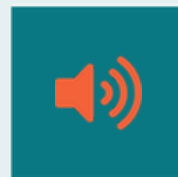
deep-translator

Wraps Google Translate & other engines to translate text between 100+ languages



SpeechRecognition + PyAudio

Captures microphone input and converts speech to text



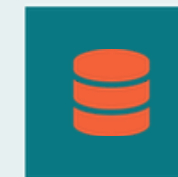
gTTS / pyttsx3

Converts translated text into natural spoken audio



Tkinter / Streamlit

Provides a simple, interactive graphical user interface



Pandas (optional)

Logs conversation history and language usage statistics

How It Works — Step by Step

01

Capture Input

User speaks into the mic or types a sentence in any language.

02

Convert Speech to Text

If voice input, SpeechRecognition converts audio to text.

03

Detect Language

langdetect analyzes the text and returns the source language code.

04

Translate Text

deep-translator sends text to the translation engine and receives translated output.

05

Speak / Display Result

gTTS converts the result to audio, and the GUI displays both text and speech.

Core Modules



Language Detection Module

Accepts raw text (typed or transcribed from speech)

Uses n-gram based statistical models (langdetect / langid) to identify the language

Returns a language code (e.g. 'hi', 'en', 'fr') with a confidence score

Works offline — no internet dependency for detection



Translation & Speech Module

- Sends detected text + target language to deep-translator (Google Translate backend)
- Supports 100+ languages for text translation
- gTTS converts translated text into an MP3 audio response
- pyttsx3 available as an offline fallback for text-to-speech

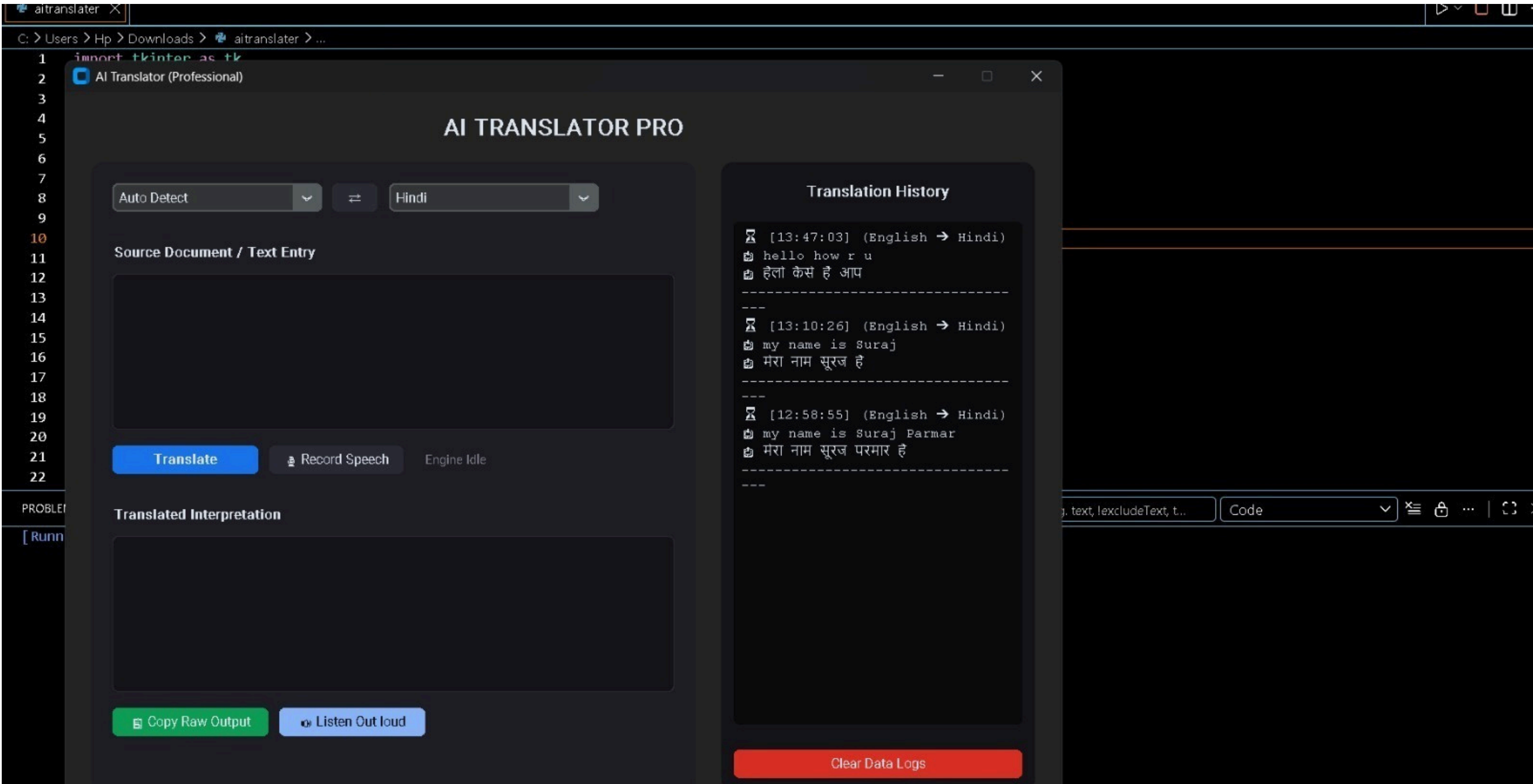
Sample Python Implementation



```
from langdetect import detectfrom deep_translator import GoogleTranslator
from gtts import gTTS

def translate_realtime(text, target_lang='en'):
    source_lang = detect(text)
    translated = GoogleTranslator(source=source_lang,
                                  target=target_lang).translate(text)
    speech = gTTS(text=translated, lang=target_lang)
    speech.save('output.mp3')
    return source_lang, translated

# Example usage
translate_realtime("Bonjour tout le monde", target_lang='hi')
```



Key Features & Advantages



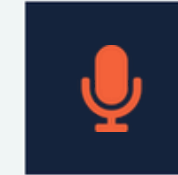
Real-Time Performance

Detection and translation complete in under a second for smooth, chat-like interaction.



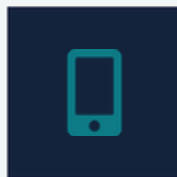
Multi-Language Support

Supports 100+ languages including Hindi, English, French, Spanish, Chinese and more.



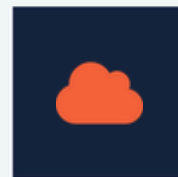
Voice & Text Input

Works with both typed sentences and spoken audio captured from a microphone.



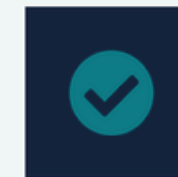
Simple GUI

Clean Tkinter / Streamlit interface — no technical knowledge needed to use it.



Lightweight & Low-Cost

Built on free, open-source Python libraries — no expensive licensing required.



Accurate Detection

Statistical language models give high accuracy even on short text snippets.

Real-World Applications



Travel & Tourism

Helps tourists communicate with locals in unfamiliar languages instantly.



Customer Support

Enables support agents to chat with customers across different countries.



Education

Assists students in understanding foreign-language study material and lectures.



Business Communication

Speeds up cross-border meetings, emails and negotiations.



Healthcare

Bridges the gap between doctors and patients who speak different languages.



Social Media & Chat Apps

Can be integrated as a plugin for instant messaging platforms.

Future Scope



Offline Neural Translation

Integrate on-device transformer models (e.g. MarianMT, NLLB) to remove internet dependency.



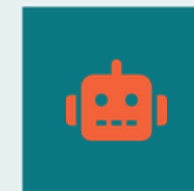
Mobile Application

Package the system into an Android/iOS app for on-the-go translation.



Chatbot Integration

Embed the translator into chatbots and virtual assistants for multilingual support.



Accent & Dialect Handling

Improve speech recognition accuracy across regional accents and dialects.

Conclusion

- The Real-Time Language Detection and AI Translator System successfully demonstrates how Python's NLP and speech libraries can be combined to break language barriers.
- By automating language detection and translation, the system saves time and effort compared to manual lookup-based translation tools.
- With voice input, instant translation and spoken output, it offers a complete, accessible communication solution for everyday use.



Thank You!

Questions & Suggestions Are Welcome